

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284246

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

OIKAWA; Shuto et al.

TIMEPIECE, DISPLAY CONTROL METHOD, AND STORAGE MEDIUM

Abstract

A timepiece includes a first display including a hand, a second display, a solar panel provided at a position having a possibility that external light is blocked by the hand that is displaying time, and at least one processor that controls an operation mode of the first display and the second display. The at least one processor switches the operation mode between a first mode in which the hand of the first display displays the time and a second mode in which the hand is evacuated to a predetermined evacuation position that makes the external light that is blocked by the hand less as compared to in the first mode and the second display displays the time.

Inventors: OIKAWA; Shuto (Tokyo, JP), YOSHIDA; Kotaro (Tokyo, JP)

Applicant: CASIO COMPUTER CO., LTD. (Tokyo, JP)

Family ID: 96882691

Appl. No.: 19/069788

Filed: March 04, 2025

Foreign Application Priority Data

JP	2024-032601	Mar. 05, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: G04C10/02 (20060101); G04C3/00 (20060101); G04C3/14 (20060101); G04C13/04 (20060101); G04C17/02 (20060101)

U.S. Cl.:

CPC G04C10/02 (20130101); G04C3/008 (20130101); G04C3/14 (20130101); G04C13/0409 (20130101); G04C17/02 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2024-032601 filed on Mar. 5, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a timepiece, a display control method, and a storage medium.

DESCRIPTION OF RELATED ART

[0003] As disclosed in JP 2003-57366 A, there is a timepiece in which a secondary battery is chargeable by electric power generated by a solar panel. In this kind of timepiece, if its hands, such as an hour hand and a minute hand, cast shadows on the solar panel, the power generation efficiency decreases. To deal with this, there is known a technique of, for example, in a case where the remaining life of the secondary battery is low, evacuating the hands to a position that makes the power generation efficiency of the solar panel less likely to decrease, thereby efficiently charging the secondary battery.

SUMMARY OF THE INVENTION

[0004] According to an aspect of the present disclosure, there is provided a timepiece including:
[0005] a first display including a hand; [0006] a second display; [0007] a solar panel provided at a position having a possibility that external light is blocked by the hand that is displaying time; and [0008] at least one processor that controls an operation mode of the first display and the second display, [0009] wherein the at least one processor switches the operation mode between a first mode in which the hand of the first display displays the time and a second mode in which the hand is evacuated to a predetermined evacuation position that makes the external light that is blocked by the hand less as compared to in the first mode and the second display displays the time.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 is a block diagram illustrating a functional configuration of an electronic timepiece.
[0011] FIG. 2 is a plan view of the electronic timepiece to illustrate a first display, a second display, a third display and so forth.
[0012] FIG. 3 is a schematic sectional view of the electronic timepiece taken along line A-A in FIG. 2.
[0013] FIG. 4 illustrates the electronic timepiece operating in a second mode.
[0014] FIG. 5 is a flowchart of a control procedure for an operation mode control process.
[0015] FIG. 6 illustrates a method for determining an evacuation position in a first modification.
[0016] FIG. 7 illustrates an example of temporal change in electromotive force of a solar panel while hands are making a round.

[0017] FIG. **8** illustrates the electronic timepiece according to a second modification, the electronic timepiece operating in the second mode.

[0018] FIG. **9** is a front view of the electronic timepiece according to a second embodiment.

[0019] FIG. **10** is a front view of the electronic timepiece according to the second embodiment, the electronic timepiece operating in the second mode.

[0020] FIG. **11** is a front view of another example of the electronic timepiece according to the second embodiment.

[0021] FIG. **12** is a front view of the example of the electronic timepiece according to the second embodiment, the electronic timepiece operating in the second mode.

DETAILED DESCRIPTION

[0022] Hereinafter, one or more embodiments of the present disclosure will be described with reference to the drawings.

First Embodiment

<Configuration of Electronic Timepiece>

[0023] FIG. **1** is a block diagram illustrating a functional configuration of an electronic timepiece **1** (timepiece). The electronic timepiece **1** is a wristwatch (wearable device) that is worn on a wrist of a user with a band (not illustrated). The electronic timepiece **1** includes a first display **10**, a second display **20**, a third display **30**, a central processing unit (CPU) **41** (at least one processor), a random access memory (RAM) **42**, a storage **43**, a time measurer **50**, an operation receiver **61**, a motion sensor **62**, a solar panel **71**, a secondary battery **72**, and a charging controller **73**. These components of the electronic timepiece **1** are connected with one another via a communication path, such as a bus, and operate by electric power supplied from the secondary battery **72**.

[0024] FIG. **2** is a plan view of the electric timepiece **1** to illustrate the first display **10**, the second display **20**, the third display **30** and so forth. The first display **10** includes an hour hand **11**, a minute hand **12** and a second hand **13** that rotate around a rotation shaft **101** at the center of a circular dial **81**. Hereinafter, the hour hand **11**, the minute hand **12** and the second hand **13** may be referred to as “hands **11** to **13**”. The first display **10** displays time in an analog form with the hands **11** to **13**. The second display **20** is disposed at six o'clock (6:00) side of the rotation shaft **101**. The second display **20** includes a segmented liquid crystal panel, and displays information such as the time, a date and a day of the week in a digital form. The third display **30** is disposed at the upper right from (1:30 side of) the rotation shaft **101**. The third display **30** includes a segmented liquid crystal panel, and displays second segments **31** indicating a lapse of seconds, an evacuation position mark **32** (in FIG. **4**), which will be described later, and so forth.

[0025] FIG. **3** is a schematic sectional view of the electronic timepiece **1** taken along line A-A in FIG. **2**. The components illustrated in FIG. **3** are housed in a case (not illustrated). The case is opened on the display surface side, and this opening is sealed with a transparent crystal (not illustrated). The electronic timepiece **1** includes a module **82** in which components for operating the first display **10**, the second display **20** and the third display **30** are housed. The hands **11** to **13** rotate independently of one another in response to rotations of the rotation shaft **101** attached to the module **82**. On the surface of the module **82** facing the hands **11** to **13** (display surface side), the solar panel **71** and the dial **81** are laid in this order. The dial **81** includes decorative members, such as indexes **811** illustrated in FIG. **2**, and the part other than the decorative members is a light transmitting part **810** made of a material having optical transparency (semi-transparent material). External light having passed through the light transmitting part **810** of the dial **81** is incident on the solar panel **71**. The decorative members may also be made of a semi-transparent material.

[0026] Hereinafter, viewing the electronic timepiece **1** in a direction perpendicular to the solar panel **71** (extension direction of the rotation shaft **101**) may be referred to as “in plan view”. As illustrated in FIG. **3**, the solar panel **71** and the dial **81** each have an opening in a region where the second display **20** is provided in plan view. Although not illustrated in FIG. **3**, the solar panel **71** and the dial **81** each also have an opening in a region where the third display **30** is provided in plan

view. That is, the solar panel **71** is provided in a region where neither the second display **20** nor the third display **30** is provided in plan view. In FIG. **2**, the region where the solar panel **71** is provided is shown in color. Since the hands **11** to **13** rotate on the solar panel **71**, the hands **11** to **13** that are displaying the time might block the external light from being incident on the solar panel **71**. Thus, the solar panel **71** is provided at a position having a possibility that the external light is blocked by the hands **11** to **13** that are displaying the time. In other words, the solar panel **71** is provided at a position having a possibility that the hands **11** to **13** cast shadows on part of the solar panel **71** in plan view.

[0027] As illustrated in FIG. **1**, the first display **10** further includes gear train mechanisms **121**, **122**, **123** that are gear trains connected to the hour hand **11**, the minute hand **12** and the second hand **13**, respectively, stepping motors **111**, **112**, **113** that rotate the gear train mechanisms **121**, **122**, **123**, respectively, and a motor drive circuit **110** that drives the stepping motors **111**, **112**, **113**. The hour hand **11** rotates angle by angle, the angle corresponding to one second, in response to the stepping of the stepping motor **111** transmitted thereto via the gear train mechanism **121**. The minute hand **12** rotates angle by angle, the angle corresponding to one second, in response to the stepping of the stepping motor **112** transmitted thereto via the gear train mechanism **122**. The second hand **13** rotates angle by angle, the angle corresponding to one second, in response to the stepping of the stepping motor **113** transmitted thereto via the gear train mechanism **123**. The rotation angle of the second hand **13** corresponding to one second is six degrees. The rotation angle of the minute hand **12** corresponding to one second is 1/60 of the rotation angle of the second hand **13**. The rotation angle of the hour hand **11** corresponding to one second is 1/12 of the rotation angle of the minute hand **12**.

[0028] The stepping motors **111** to **113** each are step-driven on the basis of the voltage waveform of a drive pulse input from the motor drive circuit **110** to rotate and move their corresponding hands **11** to **13** in the normal direction (clockwise direction in which the time advances) or the reverse direction (counterclockwise direction in which the time goes back) by their respective predetermined rotation angles mentioned above. During the time display, the stepping motors **111** to **113** are driven by a drive pulse of 1 pulse per second (pps). The stepping motors **111** to **113** can be driven by a drive pulse of up to about several tens to several hundreds pps in the normal direction and the reverse direction to rotate the hands **11** to **13** in the normal direction or the reverse direction in a fast-forward manner. Since the hour hand **11**, the minute hand **12** and the second hand **13** are connected to their respective gear train mechanisms and stepping motors, they can rotate independently of one another. In accordance with a control signal input from the CPU **41**, the motor drive circuit **110** outputs, at an appropriate timing and with an appropriate pulse width, a drive voltage pulse for driving each of the stepping motors **111** to **113** to perform the stepping.

[0029] The second display **20** and the third display **30** each include a liquid crystal panel and a drive circuit for the liquid crystal panel. The second display **20** and the third display **30** each perform the aforementioned segment display on the liquid crystal panel in accordance with a control signal and image data transmitted from the CPU **41** to the drive circuit.

[0030] The CPU **41** is a processor that functions as a controller that controls operation of the electronic timepiece **1** by reading and executing programs **431** stored in the storage **43** and performing various types of arithmetic processing accordingly. The electronic timepiece **1** may have a plurality of processors (e.g., CPUs), and processes that are performed by the CPU **41** of this embodiment may be performed by the processors. In this case, the processors constitute the controller. The processors may be involved in the same process(es), or independently perform different processes in parallel. The RAM **42** provides a working memory space for the CPU **41** and stores temporary data.

[0031] The storage **43** is a non-transitory storage medium readable by the CPU **41** as a computer, and stores the programs **431** and various data, such as setting data **432**. The storage **43** includes a nonvolatile memory, such as a flash memory. The programs **431** are stored in the storage **43** in the

form of computer-readable program codes. In the setting data **432**, data regarding various operation settings of the electronic timepiece **1** are stored.

[0032] The time measurer **50** includes an oscillator circuit, a frequency divider circuit and a time measurement circuit. In the time measurer **50**, the frequency divider circuit divides the frequency of a clock signal generated by the oscillator circuit, and the time measurement circuit counts the frequency-divided signals. Thus, the time measurer **50** derives and keeps the current date and time.

[0033] The operation receiver **61** includes operation elements (not illustrated), such as operation buttons and a crown, and outputs operation signals corresponding to operations on the operation elements to the CPU **41**.

[0034] The motion sensor **62** includes a three-axis acceleration sensor and a three-axis angular velocity sensor, and detects the acceleration and the angular velocity of the electronic timepiece **1** that are generated according to the movement of the user's wrist. On the basis of the acceleration and the angular velocity detected by the motion sensor **62**, the CPU **41** determines whether the electronic timepiece **1** is worn by the user.

[0035] The solar panel **71** includes solar cells (not illustrated) connected in series. Each solar cell generates electromotive force due to the photoelectric effect according to the external light that is incident thereon. The solar panel **71** outputs the sum of the amounts of the electromotive force of the solar cells. The solar cells may have approximate fan shapes into which the circle formed by the dial **81** illustrated in FIG. **2** is divided by separator lines extending in the radial direction, or may form a whirlpool shape with the position of the rotation shaft **101** as its center/start point, for example. The solar cells may have shapes different from one another so that the solar panel **71** is provided in the region where neither the second display **20** nor the third display **30** is provided.

[0036] The secondary battery **72** is a battery that is chargeable and dischargeable. The secondary battery **72** is charged by electric current generated by the electromotive force of the solar panel **71** flowing thereinto.

[0037] The charging controller **73** is electrically connected to the solar panel **71** and the secondary battery **72**. In accordance with a control signal transmitted from the CPU **41**, the charging controller **73** controls the charging that is performed by the solar panel **71** on the secondary battery **72**. Also, in accordance with a control signal transmitted from the CPU **41**, the charging controller **73** detects the electromotive force of the solar panel **71** and transmits the detection result to the CPU **41**. The charging controller **73** may be capable of detecting the remaining life of the secondary battery **72**.

[0038] The electronic timepiece **1** may include components not illustrated in FIG. **1**, such as a communicator that communicates with external devices and a position information obtainer that receives and decodes radio waves transmitted from positioning satellites to calculate the current position.

<Operation of Electronic Timepiece>

[0039] Next, the operation of the electronic timepiece **1** will be described. As described above, in the electronic timepiece **1** of this embodiment, the rotating hands **11** to **13** might be laid on the solar panel **71** in plan view. If the hands **11** to **13** block the external light and cast shadows on the solar panel **71**, the power generation efficiency and the electromotive force of the solar panel **71** decrease, and the charge efficiency of the secondary battery **72** decreases accordingly. Like this embodiment, in the model having the analog first display **10** and the digital second and third displays **20**, **30** in combination, the solar panel **71** cannot be provided in the regions of the second display **20** and the third display **30**. If the solar panel **71** having a limited area is shadowed by the hands **11** to **13**, the power generation efficiency is likely to decrease. This problem is more considerable if the hour hand **11** and the minute hand **12** are designed to be wide and large.

[0040] For this reason, in the electronic timepiece **1** of this embodiment, the operation mode of the first display **10**, the second display **20** and the third display **30** is switchable between a normal first mode and a second mode that is for reducing a decrease in the power generation efficiency of the

solar panel **71**. The switching of the operation mode is controlled by the CPU **41**.

[0041] The first mode is a normal mode for displaying the time. In the first mode, as illustrated in FIG. **2**, the CPU **41** causes the first display **10** to display the time (hours, minutes, seconds), causes the second display **20** to display a date and a day of the week, and causes the third display **30** to display the second segments **31**. The information displayed by the second display **20** in the first mode is not limited to a date and a day of the week, but any information other than the time.

[0042] During the operation in the first mode, if a predetermined change condition to the second mode is satisfied, the CPU **41** changes the operation mode to the second mode. The change condition to the second mode is at least one of the following: the user makes an operation indicating a change to the second mode; it is a predetermined time period (e.g., daytime) in which a high power generation efficiency is obtained; on the basis of the detection result of the motion sensor **62**, removal of the electronic timepiece **1** from the user's wrist is detected; strong light is detected by the solar panel **71** (electromotive force is a predetermined upper limit or more); and the remaining life of the secondary battery **72** is less than a predetermined first reference amount. In the second mode, as illustrated in FIG. **4**, the CPU **41** evacuates the hands **11** to **13** to and makes them still at a predetermined evacuation position that makes the external light that is blocked by the hands **11** to **13** less than that in the first mode. Also, in the second mode, the CPU **41** causes the second display **20** to display the time digitally. Also, in the second mode, the CPU **41** causes the third display **30** to display the evacuation position mark **32** in addition to the second segments **31**. The evacuation position mark **32** in this embodiment is formed of a letter of "S" indicating that the solar panel **71** is performing the charging.

[0043] As illustrated in FIG. **4**, the CPU **41** evacuates the hands **11** to **13** to the same evacuation position so that the hands **11** to **13** are laid on top of one another. At this evacuation position, the hands **11** to **13** each are partly laid on the third display **30**. The evacuation position predetermined is a position at which the hands **11** to **13** are not laid on the second display **20** and the area of a part of the solar panel **71** is the smallest, the part where the external light is blocked by the hands **11** to **13**. In other words, the evacuation position predetermined is a position at which the area of an overlap of the third display **30** and the hands **11** to **13** as viewed in the direction perpendicular to the solar panel **71** is the largest. This makes, of the external light to be incident on the solar panel **71**, the external light that is blocked by the hands **11** to **13** less than that in the first mode. Evacuating the hands **11** to **13** to this evacuation position can maximize the power generation efficiency of the solar panel **71** without interference with the time display by the second display **20**. The evacuation position mark **32** is displayed at a position to which the hands **11** to **13** at the evacuation position point. In other words, in the second mode, the CPU **41** causes the hands **11** to **13** to point to the evacuation position mark **32**.

[0044] During the operation in the second mode, if a predetermined change condition to the first mode is satisfied, the CPU **41** changes the operation mode to the first mode. The change condition to the first mode is at least one of the following: the user makes an operation indicating a change to the first mode; it is a predetermined time period (e.g., nighttime) in which the power generation efficiency decreases; on the basis of the detection result of the motion sensor **62**, wearing of the electronic timepiece **1** on the user's wrist is detected; less strong light is detected by the solar panel **71** (electromotive force is less than a predetermined lower limit); and the remaining life of the secondary battery **72** is a predetermined second reference amount or more. The second reference amount may be the same as or different from the first reference amount.

[0045] Next, an operation mode control process that is performed by the CPU **41** to realize the above-described operation of the electronic timepiece **1** will be described. FIG. **5** is a flowchart of a control procedure for the operation mode control process. The operation mode control process is started at the time of startup of the electronic timepiece **1**. When the operation mode control process is started, the CPU **41** starts a change process to the first mode (Step **S1**). That is, the CPU **41** transmits a control signal to the motor drive circuit **110** to cause the stepping motors **111** to **113**

to operate, thereby causing the hands **11** to **13** of the first display **10** to display the current time being measured by the time measurer **50** (Step S2). The CPU **41** also transmits a control signal to the drive circuit of the second display **20** to cause the second display **20** to display a date and a day of the week (Step S3). The CPU **41** also transmits a control signal to the drive circuit of the third display **30** to cause the third display **30** to display the second segment(s) **31** and not display the evacuation position mark **32** (Step S4).

[0046] The CPU **41** repeatedly determines whether the above-described change condition to the second mode is satisfied (Step S5). If the CPU **41** determines that the change condition is satisfied (Step S5; YES), the CPU **41** starts a change process to the second mode (Step S6). That is, the CPU **41** moves the hands **11** to **13** of the first display **10** to the evacuation position (Step S7). In Step S7, the CPU **41** obtains information on the evacuation position by referring to the setting data **432** and transmits a control signal to the motor drive circuit **110** to cause the stepping motors **111** to **113** to operate, thereby rotating the hour hand **11**, the minute hand **12** and the second hand **13** independently of one another in the normal direction or the reverse direction to the evacuation position in the fast-forward manner. The CPU **41** also transmits a control signal to the drive circuit of the second display **20** to cause the second display **20** to display the time (Step S8). The CPU **41** also transmits a control signal to the drive circuit of the third display **30** to cause the third display **30** to display the evacuation position mark **32** (Step S9).

[0047] The CPU **41** repeatedly determines whether the above-described change condition to the first mode is satisfied (Step S10). If the CPU **41** determines that the change condition is satisfied (Step S10; YES), the CPU **41** determines whether an operation to power off the electronic timepiece **1** has been made (Step S11). If the CPU **41** determines that the operation has not been made (Step S11; NO), the CPU **41** returns the process to Step S1, whereas if the CPU **41** determines that the operation has been made (Step S11; YES), the CPU **41** ends the operation mode control process. In a case where no power-off operation is set, the CPU **41** may return the process to Step S1 after “Step S10; YES”. That is, the loop process of Steps S1 to S10 may be repeatedly performed until the power of the secondary battery **72** runs out.

First Modification

[0048] Next, a first modification of the first embodiment will be described. Hereinafter, different points between this modification and the first embodiment will be described. In the above embodiment, the evacuation position in the second mode is preset and registered in the setting data **432**, but in this modification, at the time of the change to the second mode, an evacuation position suitable to obtain a high power generation efficiency is determined. In this modification, as an example, the electronic timepiece **1** does not have the third display **30**.

[0049] FIG. **6** illustrates a method for determining the evacuation position in the first modification. In this modification, when the operation mode is changed to the second mode, the CPU **41** causes the hands **11** to **13** to make a round in a state in which the hands **11** to **13** are laid on top of one another, and obtains, from the charging controller **73**, temporal change in the electromotive force of the solar panel **71**. The speed at which the hands **11** to **13** make a round can be appropriately determined according to a response speed at which the electromotive force of the solar panel **71** changes according to the external light incident on the solar panel **71**. This speed may be several times as fast as the normal rotation speed of the second hand **13**, for example. Of the hands **11** to **13**, only the hour hand **11** and the minute hand **12** may be caused to make a round, and the second hand **13** may be caused to continue its normal time display. This is because the second hand **13**, which is thinner than the minute hand **12**, has only a little influence on the power generation efficiency of the solar panel **71**.

[0050] FIG. **7** illustrates an example of the temporal change in the electromotive force of the solar panel **71** while the hands **11** to **13** are making a round. The horizontal axis in FIG. **7** represents the rotation angle of the hands **11** to **13** with twelve o'clock as **0°**. An angle range R represents a range of angles at which the hands **11** to **13** are laid on part of the second display **20**. As illustrated in

FIG. 7, the amounts of the electromotive force at angles are not necessarily the same. For example, if the solar panel **71** is shadowed by an object other than the hands **11** to **13**, the electromotive force is higher in a state in which the hands **11** to **13** are in the shadow than in a state in which the hands **11** to **13** are not in the shadow. Examples of the situation in which the solar panel **71** is shadowed by an object other than the hands **11** to **13** include a situation in which the external light is incident on the solar panel **71** in a direction inclined from the direction perpendicular to the solar panel **71**, so that the side wall of the case of the electronic timepiece **1** and/or the bezel thereof cast shadows on the solar panel **71**, and a situation in which the electronic timepiece **1** is at the border between a sunny place and a shady place.

[0051] On the basis of the obtained temporal change in the electromotive force, the CPU **41** determines, as the evacuation position, a position of the hands **11** to **13** at which the magnitude of the electromotive force satisfies a predetermined condition. Examples of the predetermined condition include a condition that the electromotive force at is the largest outside the angle range **R** and a condition the electromotive force is a predetermined reference value or more outside the angle range **R**. In the example illustrated in FIG. 7, the CPU **41** identifies, from among angles outside the angle range **R**, an angle **A** at which the electromotive force of the solar panel **71** is the largest, and determines a position corresponding to the identified angle **A** as the evacuation position. In this modification, the hands **11** to **13** evacuated to the evacuation position cannot point to an evacuation position mark. Therefore, the CPU **41** causes the second display **20** to display an evacuation position mark **23**. Alternatively, a function hand (not illustrated) may be provided to point to an evacuation position mark. The process of determining the evacuation position with the above-described method and evacuating the hands **11** to **13** thereto may be performed only once when the operation mode is changed to the second mode, or performed repeatedly at a predetermined frequency (e.g., once per hour) during the operation in the second mode.

Second Modification

[0052] Next, a second modification of the first embodiment will be described. Hereinafter, different points between this modification and the first embodiment will be described. The second modification may be combined with the first modification. In this modification, as an example, the electronic timepiece **1** does not have the third display **30**. In the above embodiment, the evacuation position predetermined is the position at which the hands **11** to **13** are not laid on the second display **20** but laid on the third display **30**. However, for example, in a case where the third display **30** is not provided and the solar panel **71** is provided in the entire region excluding the region of the second display **20**, it is preferable from the perspective of the reduction of the decrease in the power generation efficiency that the evacuation position predetermined be a position at which the hands **11** to **13** are laid on the second display **20**. This modification may be applied to a case too where the region excluding the region of the second display **20** includes a region where the solar panel **71** is not provided.

[0053] FIG. 8 illustrates the electronic timepiece **1** according to the second modification, the electronic timepiece **1** operating in the second mode. As illustrated in FIG. 8, in this modification, the evacuation position predetermined is a position at which parts of the hands **11** to **13** are laid on a part of the second display **20**. The CPU **41** causes the second display **20** to display the time in, of the second display **20**, a region on which the hands **11** to **13** are not laid. The second display **20** of this modification has a dot-matrix liquid crystal panel in which pixels are arranged in a matrix. The CPU **41** identifies, in the second display **20**, a region of pixels on which the hands **11** to **13** are not laid, and transmits a control signal and image data to the second display **20** for displaying the time in the region. In FIG. 8, the CPU **41** causes the second display **20** to display the evacuation position mark **23**. Alternatively, a function hand (not illustrated) may be provided to point to an evacuation position mark.

Advantageous Effects

[0054] As described above, there is known a technique of, in a timepiece in which a secondary

battery is chargeable by electric power generated by a solar panel, evacuating hands of the timepiece to a position at which the power generation efficiency of the solar panel is less likely to decrease, thereby efficiently charging the secondary battery. However, this technique has a problem that the time cannot be displayed during the period in which the hands are evacuated. To deal with this, the electronic timepiece **1** according to the first embodiment of the present disclosure is configured as follows and can obtain the following advantageous effects. The electronic timepiece **1** according to the first embodiment includes the first display **10** including the hands **11** to **13** (hand), the second display **20**, the solar panel **71** provided at a position having a possibility that the external light is blocked by the hands **11** to **13** that are displaying the time, and the CPU **41** (at least one processor) that controls the operation mode of the first display **10** and the second display **20**. The CPU **41** switches the operation mode between the first mode in which the hands **11** to **13** of the first display **10** display the time and the second mode in which the hands **11** to **13** are evacuated to a predetermined evacuation position that makes the external light that is blocked by the hands **11** to **13** less as compared to in the first mode and the second display **20** displays the time. Thus, in the second mode, the hands **11** to **13** are evacuated to the evacuation position, which can reduce the decrease in the power generation efficiency of the solar panel **71**, and also the second display **20** can display the time. Further, since the electronic timepiece **1** can operate in the second mode, the amount of power generated per day and the amount of charge of the secondary battery **72** per day can be increased, so that functions conventionally unable to be provided due to a limitation on power consumption can be added. Further, decrease in size of the electronic timepiece **1**, increase in areas of the second display **20** and the third display **30**, and increase in widths of the hands **11** to **13** are available, and the degree of freedom in design or plan can be enhanced accordingly.

[0055] Further, the hands **11** to **13** include the hour hand **11** and the minute hand **12**, and in the second mode, the CPU **41** evacuates the hour hand **11** and the minute hand **12** to the same evacuation position so that the hour hand **11** and the minute hand **12** are laid on top of one another. This can minimize the area of the shadow made by the hour hand **11** and the minute hand **12** and more effectively reduce the decrease in the power generation efficiency of the solar panel **71** accordingly.

[0056] Further, in the first modification, the CPU **41** causes the hour hand **11** and the minute hand **12** (and the second hand **13**) that are laid on top of one another to make a round, obtains the temporal change in the electromotive force generated by the solar panel **71**, and determines a position of the hour hand **11** and the minute hand **12**, the position at which the magnitude of the electromotive force satisfies a predetermined condition, as the evacuation position. Thus, the hour hand **11** and the minute hand **12** can be evacuated to the optimum evacuation position for effectively reducing the decrease in the power generation efficiency, for example, in a case where light strikes only part of the dial **81** or in a case where light strikes the dial **81** at an angle.

[0057] Further, in the second mode, the CPU **41** causes the hands **11** to **13** to point to the evacuation position mark **32** indicating that the electronic timepiece **1** is operating in the second mode. This makes it unlikely that the user misunderstands that the first display **10** is displaying the time in the second mode.

[0058] Further, the electronic timepiece **1** further includes the third display **30**, the solar panel **71** is provided in the region where the third display **30** is not provided as viewed in the direction perpendicular to the solar panel **71**, and in the second mode, the CPU **41** evacuates the hands **11** to **13** to the evacuation position at which parts of the hands **11** to **13** are laid on the third display **30**. This can effectively reduce the area of an overlap of the hands **11** to **13** and the solar panel **71**.

[0059] Further, in the second mode, the CPU **41** causes the third display **30** to display the evacuation position mark **32** indicating that the electronic timepiece **1** is operating in the second mode, and causes the hands **11** to **13** to point to the evacuation position mark **32**. This makes it unlikely that the user misunderstands that the first display **10** is displaying the time in the second mode. Further, since the evacuation position mark **32** is displayed only in the second mode, the

design of the electronic timepiece **1** in the first mode can be maintained.

[0060] Further, the solar panel **71** is provided at least in a part of the region where the second display **20** is not provided as viewed in the direction perpendicular to the solar panel **71**, and the evacuation position is a position at which the hands **11** to **13** are not laid on the second display **20** and the area of a part of the solar panel **71** is the smallest, the part where the external light is blocked by the hands **11** to **13**. This can most effectively reduce the area of the overlap of the hands **11** to **13** and the solar panel **71** without interference with the time display by the second display **20**. [0061] Further, in the second modification, in the second mode, the CPU **41** evacuates the hands **11** to **13** to the evacuation position at which parts of the hands **11** to **13** are laid on a part of the second display **20**, and causes the second display **20** to display the time in a region of the second display **20**, the region on which the hands **11** to **13** are not laid. This can effectively reduce the area of the overlap of the hands **11** to **13** and the solar panel **71**. Further, even in the state in which the hands **11** to **13** are laid on a part of the second display **20**, the second display **20** can display the time visibly.

[0062] Further, the display control method of the electronic timepiece **1** according to this embodiment includes switching the operation mode of the first display **10** and the second display **20** between the first mode in which the hands **11** to **13** (hand) of the first display **10** display the time and the second mode in which the hands **11** to **13** are evacuated to a predetermined evacuation position that makes the external light that is blocked by the hands **11** to **13** less as compared to in the first mode and the second display **20** displays the time. Thus, in the second mode, the hands **11** to **13** are evacuated to the evacuation position, which can reduce the decrease in the power generation efficiency of the solar panel **71**, and also the second display **20** can display the time.

[0063] Further, the storage **43** as the non-transitory computer-readable storage medium according to this embodiment stores the program(s) **431** executable by the CPU **41**. The program **431** causes the CPU **41** to switch the operation mode of the first display **10** and the second display **20** between the first mode in which the hands **11** to **13** (hand) of the first display **10** display the time and the second mode in which the hands **11** to **13** are evacuated to a predetermined evacuation position that makes the external light that is blocked by the hands **11** to **13** less as compared to in the first mode and the second display **20** displays the time. Thus, in the second mode, the hour hands **11** to **13** are evacuated to the evacuation position, which can reduce the decrease in the power generation efficiency of the solar panel **71**, and also the second display **20** can display the time.

Second Embodiment

[0064] Next, a second embodiment will be described. Hereinafter, different points from the first embodiment will be described, and descriptions of components in common with the first embodiment will be omitted. The second embodiment may be combined with one or both of the first modification and the second modification of the first embodiment.

[0065] FIG. **9** is a front view of the electronic timepiece **1** according to the second embodiment, the electronic timepiece **1** operating in the first mode. The second display **20** of this embodiment includes a sub hour hand **21** and a sub minute hand **22**, and displays the time with the sub hour hand **21** and the sub minute hand **22**. For example, while the first display **10** displays the current time, the second display **20** displays the local time (world time) of a preset city in the world. The sub hour hand **21** has an area smaller than that of the hour hand **11**, and the sub minute hand **22** has an area smaller than that of the minute hand **12**. The areas of the hour hand **11**, the minute hand **12**, the sub hour hand **21** and the sub minute hand **22** are the areas of regions occupied by the respective hands as viewed in the direction perpendicular to the solar panel **71**.

[0066] The third display **30** includes a function hand **33**, and displays the operation mode and so forth by pointing to various marks with the function hand **33**. The marks to which the function hand **33** points include “SW” indicating that the electronic timepiece **1** is operating in a stopwatch mode, “WT” indicating that the electronic timepiece **1** is operating in a world time display mode, “TM” indicating that the electronic timepiece **1** is operating in a timer mode, and “S” (in FIG. **10**)

indicating that the electronic timepiece **1** is operating in the above-described second mode. In this embodiment, the mark “S” corresponds to the evacuation position mark **32**.

[0067] The second display **20** is circular and disposed at six o'clock side of the rotation shaft **101**.

The third display **30** is circular and disposed at twelve o'clock side of the rotation shaft **101**.

Similarly to the first display **10** of the first embodiment, the second display **20** and the third display **30** each include a stepping motor(s) that is driven by the motor drive circuit **110** and a gear train mechanism(s) that transmits the stepping of the stepping motor(s) to the sub hour hand **21** and the sub minute hand **22** or to the function hand **33**. The sub hour hand **21** and the sub minute hand **22** may be configured to rotate in connection with one another. That is, the sub hour hand **21** and the sub minute hand **22** may be connected to a gear train mechanism for the sub hour hand **21** and a gear train mechanism for the sub minute hand **22**, respectively, that transmit the stepping of one stepping motor that is common to the sub hour hand **21** and the sub minute hand **22**.

[0068] The solar panel **71** of this embodiment is provided in regions where the circular second display **20** and the circular third display **30** are provided in plan view. The solar panel **71** provided in in-dials (second display **20** and third display **30**) is called an in-dial solar panel. The in-dial solar panel has an advantage that the design of the dial **81** located outside the second display **20** and the third display **30** can be enhanced.

[0069] FIG. **10** is a front view of the electronic timepiece **1** according to the second embodiment, the electronic timepiece **1** operating in the second mode. In this embodiment, in the second mode, the CPU **41** evacuates the hour hand **11** and the minute hand **12** of the first display **10** to the evacuation position, and causes the sub hour hand **21** and the sub minute hand **22** of the second display **20** to display the time. Since the second display **20** has no second hand, the CPU **41** rotates the second hand **13** of the first display **10** to display seconds in the same manner as in the first mode, but not limited to thereto and may evacuate all the hands **11** to **13** to the evacuation position.

[0070] Since the in-dial solar panel **71** generates the power with a small area, even if the solar panel **71** is partly shadowed by the hour hand **11** and the minute hand **12**, the power generation efficiency decreases significantly. For this reason, the evacuation position predetermined is a position at which the hour hand **11** and the minute hand **12** are laid on neither the second display **20** nor the third display **30**, namely, a position at which the hour hand **11** and the minute hand **12** are not laid on the solar panel **71**. Of the dial **81**, at a position to which the hands **11**, **12** at the evacuation position point, an evacuation position mark **812** is provided. In other words, in the second mode, the CPU **41** causes the hands **11**, **12** to point to the evacuation position mark **812**. In the second mode, the CPU **41** also causes the function hand **33** to point to the evacuation position mark **32**.

[0071] The hour hand **11** and the minute hand **12** evacuated to the evacuation position in FIG. **10** point to a position between two o'clock and three o'clock, the position being closer to three o'clock. This arrangement of the hour hand **11** and the minute hand **12** is different from any arrangement thereof that appears during the time display. This arrangement can be realized by the hour hand **11** and the minute hand **12** configured to rotate independently of one another. Evacuating the hour hand **11** and the minute hand **12** to this evacuation position allows the user to intuitively recognize that the first display **10** is not displaying the time.

[0072] FIG. **11** is a front view of another example of the electronic timepiece **1** according to the second embodiment, the electronic timepiece **1** operating in the first mode. In the electronic timepiece **1** illustrated in FIG. **11**, the dial **81** includes an annular decorative part **813** provided around the second display **20** and an annular decorative part **814** provided around the third display **30**. The decorative part **813** and the decorative part **814** are made of an opaque material and correspond to a light blocking part that blocks light. Of the dial **81**, the part excluding the decorative part **813**, the decorative part **814** and the indexes **811** is the light transmitting part **810** (in FIG. **3**) having optical transparency. The decorative part **813** has hour characters representing the time that the second display **20** displays. The decorative part **814** has marks to which the

function hand **33** of the third display **30** points.

[0073] The solar panel **71** is provided in regions where neither the decorative part **813** nor the decorative part **814** is provided as viewed in the direction perpendicular to the solar panel **71**. That is, the solar panel **71** is provided in regions that coincide with the inside of the second display **20** in the decorative part **813**, the inside of the third display **30** in the decorative part **814**, and the region of the dial **81** outside the decorative parts **813**, **814**. In this specification, the “region(s) where the solar panel **71** is provided” refers to the region in which the solar panel **71** receives light that is to enter the solar panel **71** from the display surface side in the direction perpendicular to the solar panel **71**. Therefore, the solar panel **71** may extend to be partly laid on the back surface side of the decorative parts **813**, **814**. In this form too, the “region(s) where the solar panel **71** is provided” is those colored in FIG. **12**.

[0074] FIG. **12** is a front view of the electronic timepiece **1** illustrated in FIG. **11**, the electronic timepiece **1** operating in the second mode. The predetermined evacuation position in the electronic timepiece **1** illustrated in FIG. **12** is a position at which parts of the hour hand **11** and the minute hand **12** are laid on a part of the decorative part **814**. Since the regions where the decorative parts **813**, **814** are provided do not contribute to the power generation by the solar panel **71**, the hour hand **11** and the minute hand **12** are evacuated to be laid on a part of the decorative part **814** or the decorative part **813**. This can reduce the decrease in the power generation efficiency of the solar panel **71**. Further, of the dial **81**, at the position to which the hands **11**, **12** at the evacuation position point, the evacuation position mark **812** is provided.

Advantageous Effects

[0075] As described above, the second display **20** of the electronic timepiece **1** according to the second embodiment includes the sub hour hand **21** having an area smaller than the area of the hour hand **11** and the sub minute hand **22** having an area smaller than the area of the minute hand **12**, the solar panel **71** is provided at least in the region where the second display **20** is provided as viewed in the direction perpendicular to the solar panel **71**, and in the second mode, the CPU **41** evacuates the hour hand **11** and the minute hand **12** to the evacuation position at which the hour hand **11** and the minute hand **12** are not laid on the second display **20**. Thus, in the electronic timepiece **1** using the in-dial solar panel **71**, the decrease in the power generation efficiency of the solar panel **71** can be reduced, and also the second display **20** can display the time.

[0076] Further, the hands **11** to **13** (hand) include the second hand **13** in addition to the hour hand **11** and the minute hand **12**, and in the second mode, the CPU **41** causes the second display **20** to display hours and minutes, and causes the second hand **13** of the first display **10** to display seconds. Thus, although the second display **20** has no second hand, seconds can be displayed in the second mode.

[0077] Further, the CPU **41** is capable of rotating the hour hand **11** and the minute hand **12** independently of one another, and in the second mode, evacuates the hour hand **11** and the minute hand **12** to the evacuation position that is a position at which the hour hand **11** and the minute hand **12** are not laid on top of one another while the hour hand **11** and the minute hand **12** are displaying the time. This allows the user to intuitively recognize that the first display **10** is not displaying the time, and makes it unlikely that the user misunderstands that the first display **10** is displaying the time in the second mode accordingly.

[0078] Further, the electronic timepiece **1** further includes the dial **81** provided between the hands **11** to **13** and the solar panel **71** and including the light transmitting part **810** that transmits the external light and the decorative parts **813**, **814** that block the external light, the solar panel **71** is provided in the regions where neither the decorative part **813** nor the decorative part **814** is provided as viewed in the direction perpendicular to the solar panel **71**, and in the second mode, the CPU **41** evacuates the hour hand **11** and the minute hand **12** to the evacuation position at which parts of the hour hand **11** and the minute hand **12** are laid on the decorative part **813** or **814**. Thus, the decorative parts **813**, **814** are utilized to effectively reduce the area of an overlap of the hands

11, 12 and the solar panel **71**.

Others

[0079] The present disclosure is not limited to the above embodiments and the like, and can be modified in a variety of aspects. For example, in FIG. 4 for the first embodiment and FIG. 8 for the second modification, the second hand **13** is evacuated to the evacuation position together with the hour hand **11** and the minute hand **12**, but not limited thereto. Since the second hand **13**, which is thinner than the minute hand **12**, has only a little influence on the power generation efficiency of the solar panel **71**, in the second mode, only the hour hand **11** and the minute hand **12** may be evacuated to the evacuation position, and the second hand **13** may be caused to rotate to display seconds as in the first mode. Further, instead of the hour hand **11** and the minute hand **12** configured to point to the evacuation position mark **32** or **812**, the second hand **13** may be configured to point to the evacuation position mark **32** or **812** provided or displayed at an appropriate position.

[0080] Further, in the above, the hour hand **11** and the minute hand **12** are evacuated to the same evacuation position, but if there are two or more positions usable as the evacuation position for effectively avoiding the decrease in the power generation efficiency of the solar panel **71**, the hour hand **11** and the minute hand **12** may be evacuated to different evacuation positions.

[0081] Further, in the above, the hour hand **11** and the minute hand **12** are configured to rotate independently of one another, but may be configured to rotate in connection with one another.

[0082] Further, the region where the solar panel **71** is provided is not limited to those exemplified in the above embodiments and modifications, but any region as far as it is a region having a possibility that the external light is blocked by the hour hand **11** and the minute hand **12** that are displaying the time.

[0083] Further, in the above, the electronic timepiece **1** is a wristwatch, but may be another type of timepiece, such as a table clock or a wall clock.

[0084] Further, in the above, the computer-readable medium storing the program(s) according to the present disclosure is the flash memory of the storage **43**, but not limited thereto. The computer-readable medium may be an information storage medium, such as a hard disk drive (HDD), a solid state drive (SSD) or a CD-ROM. Further, carrier waves are also applied to the present disclosure as a medium that provides data of the program(s) according to the present disclosure via a communication line.

[0085] Further, it goes without saying that the detailed configuration and operation of each component of the electronic timepiece **1** in the above embodiments and the like can be modified as appropriate without departing from the scope of the present disclosure.

[0086] Although one or more embodiments and the like of the present disclosure have been described above, the scope of the present disclosure is not limited to the above-described embodiments and the like, but includes the scope of claims below and the scope of their equivalents.

Claims

1. A timepiece comprising: a first display including a hand; a second display; a solar panel provided at a position having a possibility that external light is blocked by the hand that is displaying time; and at least one processor that controls an operation mode of the first display and the second display, wherein the at least one processor switches the operation mode between a first mode in which the hand of the first display displays the time and a second mode in which the hand is evacuated to a predetermined evacuation position that makes the external light that is blocked by the hand less as compared to in the first mode and the second display displays the time.
2. The timepiece according to claim 1, wherein the hand includes an hour hand and a minute hand, and wherein in the second mode, the at least one processor evacuates the hour hand and the minute

hand to the evacuation position so that the hour hand and the minute hand are laid on top of one another.

3. The timepiece according to claim 2, wherein the at least one processor causes the hour hand and the minute hand that are laid on top of one another to make a round, obtains temporal change in electromotive force generated by the solar panel, and determines a position of the hour hand and the minute hand, the position at which a magnitude of the electromotive force satisfies a predetermined condition, as the evacuation position.

4. The timepiece according to claim 1, wherein in the second mode, the at least one processor causes the hand to point to a predetermined mark indicating that the timepiece is operating in the second mode.

5. The timepiece according to claim 1, further comprising a third display, wherein the solar panel is provided in a region where the third display is not provided as viewed in a direction perpendicular to the solar panel, and wherein in the second mode, the at least one processor evacuates the hand to the evacuation position at which a part of the hand is laid on the third display.

6. The timepiece according to claim 5, wherein in the second mode, the at least one processor causes the third display to display a predetermined mark indicating that the timepiece is operating in the second mode, and causes the hand to point to the mark.

7. The timepiece according to claim 1, wherein the solar panel is provided at least in a part of a region where the second display is not provided as viewed in a direction perpendicular to the solar panel, and wherein the evacuation position is a position at which the hand is not laid on the second display and an area of a part of the solar panel is smallest, the part where the external light is blocked by the hand.

8. The timepiece according to claim 1, wherein in the second mode, the at least one processor evacuates the hand to the evacuation position at which a part of the hand is laid on a part of the second display, and causes the second display to display the time in a region of the second display, the region on which the hand is not laid.

9. The timepiece according to claim 2, wherein the second display includes a sub hour hand having an area smaller than an area of the hour hand of the first display and a sub minute hand having an area smaller than an area of the minute hand of the first display, wherein the solar panel is provided at least in a region where the second display is provided as viewed in a direction perpendicular to the solar panel, and wherein in the second mode, the at least one processor evacuates the hour hand and the minute hand of the first display to the evacuation position at which the hour hand and the minute hand of the first display are not laid on the second display.

10. The timepiece according to claim 2, wherein the hand further includes a second hand, and wherein in the second mode, the at least one processor causes the second display to display an hour and a minute, and causes the second hand of the first display to display a second.

11. The timepiece according to claim 2, wherein the at least one processor is capable of rotating the hour hand and the minute hand independently of one another, and in the second mode, evacuates the hour hand and the minute hand to the evacuation position that is a position at which the hour hand and the minute hand are not laid on top of one another while the hour hand and the minute hand are displaying the time.

12. The timepiece according to claim 1, further comprising a dial provided between the hand and the solar panel and including a light transmitting part that transmits the external light and a light blocking part that blocks the external light, wherein the solar panel is provided in a region where the light blocking part is not provided as viewed in a direction perpendicular to the solar panel, and wherein in the second mode, the at least one processor evacuates the hand to the evacuation position at which a part of the hand is laid on the light blocking part.

13. A display control method that is performed by a computer of a timepiece including a first display including a hand, a second display, and a solar panel provided at a position having a possibility that external light is blocked by the hand that is displaying time, the display control

method comprising: switching an operation mode of the first display and the second display between a first mode in which the hand of the first display displays the time and a second mode in which the hand is evacuated to a predetermined evacuation position that makes the external light that is blocked by the hand less as compared to in the first mode and the second display displays the time.

14. The display control method according to claim 13 wherein the hand includes an hour hand and a minute hand, and wherein in the second mode, the hour hand and the minute hand are evacuated to the evacuation position so that the hour hand and the minute hand are laid on top of one another.

15. The display control method according to claim 14, further comprising: causing the hour hand and the minute hand that are laid on top of one another to make a round; obtaining temporal change in electromotive force generated by the solar panel; and determining a position of the hour hand and the minute hand, the position at which a magnitude of the electromotive force satisfies a predetermined condition, as the evacuation position.

16. The display control method according to claim 13, wherein in the second mode, the hand is evacuated to the evacuation position at which a part of the hand is laid on a part of the second display, and the second display is caused to display the time in a region of the second display, the region on which the hand is not laid.

17. A non-transitory computer-readable storage medium storing a program causing, of a timepiece including a first display including a hand, a second display, and a solar panel provided at a position having a possibility that external light is blocked by the hand that is displaying time, a computer to: switch an operation mode of the first display and the second display between a first mode in which the hand of the first display displays the time and a second mode in which the hand is evacuated to a predetermined evacuation position that makes the external light that is blocked by the hand less as compared to in the first mode and the second display displays the time.

18. The non-transitory computer-readable storage medium according to claim 17, wherein the hand includes an hour hand and a minute hand, and wherein the program causes the computer to, in the second mode, evacuate the hour hand and the minute hand to the evacuation position so that the hour hand and the minute hand are laid on top of one another.

19. The non-transitory computer-readable storage medium according to claim 18, wherein the program further causes the computer to: cause the hour hand and the minute hand that are laid on top of one another to make a round; obtain temporal change in electromotive force generated by the solar panel; and determine a position of the hour hand and the minute hand, the position at which a magnitude of the electromotive force satisfies a predetermined condition, as the evacuation position.

20. The non-transitory computer-readable storage medium according to claim 17, wherein the program causes the computer to, in the second mode, evacuate the hand to the evacuation position at which a part of the hand is laid on a part of the second display, and cause the second display to display the time in a region of the second display, the region on which the hand is not laid.
